

Lecture 9

# Naïve Bayes; Review and Final Thoughts

DSC 40A, Summer 2026

Lecture 9 Part 1

# Naïve Bayes

DSC 40A, Summer 2026

## Agenda

- Classification.
- Classification and conditional independence.
- Naïve Bayes.

## Recap: Bayes' Theorem, independence, and conditional independence

- Bayes' Theorem describes how to update the probability of one event, given that another event has occurred.

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(B) \cdot \mathbb{P}(A|B)}{\mathbb{P}(A)}$$

Handwritten annotations: "new" with an arrow pointing to  $\mathbb{P}(B|A)$ , "posterior" with an arrow pointing to  $\mathbb{P}(B|A)$ , "old (prior)" with an arrow pointing to  $\mathbb{P}(B)$ , and "evidence" with an arrow pointing to  $\mathbb{P}(A)$ .

- $A$  and  $B$  are **independent** if:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

- $A$  and  $B$  are **conditionally independent** given  $C$  if:

$$\mathbb{P}((A \cap B)|C) = \mathbb{P}(A|C) \cdot \mathbb{P}(B|C)$$

- In general, there is no relationship between independence and conditional independence.



**Question** 🤔

**Answer in chat.**

**Remember, you can always ask questions in Q&A!**

# Classification

# Classification problems

- Like with regression, we're interested in making predictions based on data (called **training data**) for which we know the value of the response variable.
- The difference is that the response variable is now **categorical**.
- Categories are called **classes**.
- Example classification problems:
  - Deciding whether a patient has kidney disease.
  - Identifying handwritten digits.
  - Determining whether an avocado is ripe. ✓
  - Predicting whether credit card activity is fraudulent.
  - Predicting whether you'll be late to school or not.

## Example: Avocados

| color        | ripeness |
|--------------|----------|
| bright green | unripe ✗ |
| green-black  | ripe ✓   |
| purple-black | ripe ✓   |
| green-black  | unripe ✗ |
| purple-black | ripe ✓   |
| bright green | unripe ✗ |
| green-black  | ripe ✓   |
| purple-black | ripe ✓   |
| green-black  | ripe ✓   |
| green-black  | unripe ✗ |
| purple-black | ripe ✓   |

You have a green-black avocado, and want to know if it is ripe. Based on this data, would you predict your avocado is ripe or unripe?

**Strategy:** Calculate two probabilities:

$$\mathbb{P}(\text{ripe}|\text{green-black})$$

$$\frac{3}{5}$$

$$\mathbb{P}(\text{unripe}|\text{green-black})$$

$$\frac{2}{5}$$

Then, predict the class with a **larger** probability.

$$\frac{3}{5} > \frac{2}{5} \Rightarrow \text{ripe} \quad \text{H(gb)}$$

*prediction*

population parameters

## Estimating probabilities

- We would like to determine  $\mathbb{P}(\text{ripe}|\text{green-black})$  and  $\mathbb{P}(\text{unripe}|\text{green-black})$  for all avocados in the universe.
- All we have is a single dataset, which is a **sample** of all avocados in the universe.
- We can estimate these probabilities by using sample proportions. *→ Sample Statistics*

$$\mathbb{P}(\text{ripe}|\text{green-black}) \approx \frac{\# \text{ ripe green-black avocados in sample}}{\# \text{ green-black avocados in sample}}$$

- Per the **law of large numbers** in DSC 10, larger samples lead to more reliable estimates of population parameters.

## Bayes' Theorem for Classification

- Suppose that  $A$  is the event that an avocado has certain features, and  $B$  is the event that an avocado belongs to a certain class. Then, by Bayes' Theorem:

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(B) \cdot \mathbb{P}(A|B)}{\mathbb{P}(A)}$$

- More generally:

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

- What's the point?
  - Usually, it's not possible to estimate  $\mathbb{P}(\text{class}|\text{features})$  directly.
  - Instead, we often have to estimate  $\mathbb{P}(\text{class})$ ,  $\mathbb{P}(\text{features}|\text{class})$ , and  $\mathbb{P}(\text{features})$  separately.

## Example: Avocados

| color        | ripeness |
|--------------|----------|
| bright green | unripe ✗ |
| green-black  | ripe ✓   |
| purple-black | ripe ✓   |
| green-black  | unripe ✗ |
| purple-black | ripe ✓   |
| bright green | unripe ✗ |
| green-black  | ripe ✓   |
| purple-black | ripe ✓   |
| green-black  | ripe ✓   |
| green-black  | unripe ✗ |
| purple-black | ripe ✓   |

You have a green-black avocado, and want to know if it is ripe. Based on this data, would you predict your avocado is ripe or unripe?

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

**Shortcut:** Both probabilities have the same denominator, so the larger probability is the one with the **larger numerator**.

$$\begin{aligned} \mathbb{P}(\text{ripe}|\text{green-black}) &= \frac{p(\text{ripe}) p(\text{gb}|\text{ripe})}{p(\text{gb})} = \frac{\frac{7}{11} \cdot \frac{3}{7}}{\frac{5}{11}} = \frac{3}{5} \\ \mathbb{P}(\text{unripe}|\text{green-black}) &= \frac{p(\text{unripe}) p(\text{gb}|\text{unripe})}{p(\text{gb})} = \frac{\frac{4}{11} \cdot \frac{2}{11}}{\frac{5}{11}} = \frac{2}{5} \end{aligned}$$

# Classification and conditional independence



## Example: Avocados, but with more features

| color        | softness | variety | ripeness |
|--------------|----------|---------|----------|
| bright green | firm     | Zutano  | unripe   |
| green-black  | medium   | Hass    | ripe     |
| purple-black | firm     | Hass    | ripe     |
| green-black  | medium   | Hass    | unripe   |
| purple-black | soft     | Hass    | ripe     |
| bright green | firm     | Zutano  | unripe   |
| green-black  | soft     | Zutano  | ripe     |
| purple-black | soft     | Hass    | ripe     |
| green-black  | soft     | Zutano  | ripe     |
| green-black  | firm     | Hass    | unripe   |
| purple-black | medium   | Hass    | ripe     |

You have a firm green-black Zutano avocado. Based on this data, would you predict that your avocado is ripe or unripe?

**Strategy:** Calculate  $\mathbb{P}(\text{ripe}|\text{features})$  and  $\mathbb{P}(\text{unripe}|\text{features})$  and choose the class with the **larger** probability.

**Issue:** We have not seen a firm green-black Zutano avocado before, which means that the following probabilities are undefined:

$$\mathbb{P}(\text{ripe}|\text{firm, green-black, Zutano})$$

$$\mathbb{P}(\text{unripe}|\text{firm, green-black, Zutano})$$

## A simplifying assumption

- We want to find  $\mathbb{P}(\text{ripe}|\text{firm, green-black, Zutano})$ , but there are no firm green-black Zutano avocados in our dataset.
- Bayes' Theorem tells us this probability is equal to:

$$\mathbb{P}(\text{ripe}|\text{firm, green-black, Zutano}) = \frac{\mathbb{P}(\text{ripe}) \cdot \mathbb{P}(\text{firm, green-black, Zutano}|\text{ripe})}{\mathbb{P}(\text{firm, green-black, Zutano})}$$

- **Key idea:** Assume that features are **conditionally independent** given a class (e.g. ripe).

$$\mathbb{P}(\text{firm, green-black, Zutano}|\text{ripe}) = \mathbb{P}(\text{firm}|\text{ripe}) \cdot \mathbb{P}(\text{green-black}|\text{ripe}) \cdot \mathbb{P}(\text{Zutano}|\text{ripe})$$

## Example: Avocados, but with more features

| color        | softness | variety | ripeness |
|--------------|----------|---------|----------|
| bright green | firm     | Zutano  | unripe   |
| green-black  | medium   | Hass    | ripe     |
| purple-black | firm     | Hass    | ripe     |
| green-black  | medium   | Hass    | unripe   |
| purple-black | soft     | Hass    | ripe     |
| bright green | firm     | Zutano  | unripe   |
| green-black  | soft     | Zutano  | ripe     |
| purple-black | soft     | Hass    | ripe     |
| green-black  | soft     | Zutano  | ripe     |
| green-black  | firm     | Hass    | unripe   |
| purple-black | medium   | Hass    | ripe     |

You have a firm green-black Zutano avocado. Based on this data, would you predict that your avocado is ripe or unripe?

$$p(\text{ripe} | \text{firm, gb, Zutano})$$
$$p(\text{unripe} | \text{firm, gb, Zutano})$$

$$\mathbb{P}(\text{ripe}|\text{firm, green-black, Zutano}) = \frac{\mathbb{P}(\text{ripe}) \cdot \mathbb{P}(\text{firm, green-black, Zutano}|\text{ripe})}{\mathbb{P}(\text{firm, green-black, Zutano})}$$

$$\begin{aligned} & \propto \mathbb{P}(\text{ripe}) \cdot \mathbb{P}(\text{firm, green-black, Zutano}|\text{ripe}) \leftarrow \text{(conditional)} \\ & = \mathbb{P}(\text{ripe}) \mathbb{P}(\text{firm}|\text{ripe}) \mathbb{P}(\text{green-black}|\text{ripe}) \mathbb{P}(\text{Zutano}|\text{ripe}) \leftarrow \text{independent} \\ & = \left(\frac{7}{11}\right) \left(\frac{1}{7}\right) \left(\frac{3}{7}\right) \left(\frac{2}{7}\right) \leftarrow \\ & = \frac{6}{539} \end{aligned}$$

$$\begin{aligned}
\mathbb{P}(\text{unripe}|\text{firm, green-black, Zutano}) &= \frac{\mathbb{P}(\text{unripe}) \cdot \mathbb{P}(\text{firm, green-black, Zutano}|\text{unripe})}{\mathbb{P}(\text{firm, green-black, Zutano})} \\
&\propto \mathbb{P}(\text{unripe}) \cdot \mathbb{P}(\text{firm, green-black, Zutano}|\text{unripe}) \leftarrow \\
&= \mathbb{P}(\text{unripe})\mathbb{P}(\text{firm}|\text{unripe})\mathbb{P}(\text{green-black}|\text{unripe})\mathbb{P}(\text{Zutano}|\text{unripe}) \leftarrow \\
&= \left(\frac{4}{11}\right)\left(\frac{3}{4}\right)\left(\frac{2}{4}\right)\left(\frac{2}{4}\right) \leftarrow \\
&= \frac{3}{44} = \frac{6}{88} \leftarrow
\end{aligned}$$

## Conclusion

- The numerator of  $\mathbb{P}(\text{ripe}|\text{firm, green-black, Zutano})$  is  $\frac{6}{539}$ .
- The numerator of  $\mathbb{P}(\text{unripe}|\text{firm, green-black, Zutano})$  is  $\frac{6}{88}$ .
- Both probabilities have the same denominator,  $\mathbb{P}(\text{firm, green-black, Zutano})$ .
- Since we're just interested in seeing which one is larger, we can ignore the denominator and compare numerators.
- Since the numerator for unripe is **larger** than the numerator for ripe, we **predict that our avocado is unripe** ✗.

# Naïve Bayes

Goal: prediction  
(Best)

## The Naïve Bayes classifier

- We want to predict a class, given certain features.
- Using Bayes' Theorem, we write:

*rise, write*  $\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$  *↓ gb, futano* *↓ conditionally independent on class*

- For each class, we compute the numerator using the **naïve assumption of conditional independence of features given the class.**
- We estimate each term in the numerator based on the training data.
- We predict the class with the largest numerator.
  - Works if we have multiple classes, too!



# Dictionary

Definitions from [Oxford Languages](#) · [Learn more](#)



## na·ive

/nä'ēv/

*adjective*

(of a person or action) showing a lack of experience, wisdom, or judgment.

"the rather naive young man had been totally misled"

- (of a person) natural and unaffected; innocent.  
"Andy had a sweet, naive look when he smiled"

Similar:

innocent

unsophisticated

artless

ingenuous

inexperienced



- of or denoting art produced in a straightforward style that deliberately rejects sophisticated artistic techniques and has a bold directness resembling a child's work, typically in bright colors with little or no perspective.

## Example: Avocados, again

| color        | softness | variety | ripeness |
|--------------|----------|---------|----------|
| bright green | firm     | Zutano  | unripe   |
| green-black  | medium   | Hass    | ripe     |
| purple-black | firm     | Hass    | ripe     |
| green-black  | medium   | Hass    | unripe   |
| purple-black | soft     | Hass    | ripe     |
| bright green | firm     | Zutano  | unripe   |
| green-black  | soft     | Zutano  | ripe     |
| purple-black | soft     | Hass    | ripe     |
| green-black  | soft     | Zutano  | ripe     |
| green-black  | firm     | Hass    | unripe   |
| purple-black | medium   | Hass    | ripe     |

You have a soft green-black Hass avocado. Based on this data, would you predict that your avocado is ripe or unripe?

$$\begin{aligned}
 & P(\text{ripe} \mid \text{soft}, \text{gb}, \text{Hass}) \\
 & \propto P(\text{ripe}) P(\text{soft} \mid \text{ripe}) P(\text{gb} \mid \text{ripe}) P(\text{Hass} \mid \text{ripe}) \\
 & \text{some } \# = \frac{7}{11} \cdot \frac{4}{7} \cdot \frac{3}{7} \cdot \frac{5}{7}
 \end{aligned}$$

$$\begin{aligned}
 & P(\text{unripe} \mid \text{soft}, \text{gb}, \text{Hass}) \\
 & \propto P(\text{unripe}) P(\text{soft} \mid \text{unripe}) P(\text{gb} \mid \text{unripe}) P(\text{Hass} \mid \text{unripe}) \\
 & \frac{4}{11} \cdot \frac{0}{4} \cdot \frac{3}{4} \cdot \frac{2}{4} = 0
 \end{aligned}$$

## Uh oh!

- There are no soft unripe avocados in the data set.
- The estimate  $\mathbb{P}(\text{soft}|\text{unripe}) \approx \frac{\# \text{ soft unripe avocados}}{\# \text{ unripe avocados}}$  is 0.

- The estimated numerator:

$$\mathbb{P}(\text{unripe}) \cdot \mathbb{P}(\text{soft, green-black, Hass}|\text{unripe}) = \mathbb{P}(\text{unripe}) \cdot \mathbb{P}(\text{soft}|\text{unripe}) \cdot \mathbb{P}(\text{green-black}|\text{unripe}) \cdot \mathbb{P}(\text{Hass}|\text{unripe})$$

is also 0.

- But just because there isn't a soft unripe avocado in the data set, doesn't mean that it's impossible for one to exist!
- **Idea:** Adjust the numerators and denominators of our estimate so that they're never 0.

↳ if we haven't seen that condition in the dataset,  
just ignore it

# Smoothing

- Without smoothing: *feature class*

$$\begin{aligned} \mathbb{P}(\text{soft}|\text{unripe}) &\approx \frac{\# \text{ soft unripe}}{\# \text{ soft unripe} + \# \text{ medium unripe} + \# \text{ firm unripe}} \\ \mathbb{P}(\text{medium}|\text{unripe}) &\approx \frac{\# \text{ medium unripe}}{\# \text{ soft unripe} + \# \text{ medium unripe} + \# \text{ firm unripe}} \\ \mathbb{P}(\text{firm}|\text{unripe}) &\approx \frac{\# \text{ firm unripe}}{\# \text{ soft unripe} + \# \text{ medium unripe} + \# \text{ firm unripe}} \end{aligned}$$

*add to*

- With smoothing:

$$\begin{aligned} \mathbb{P}(\text{soft}|\text{unripe}) &\approx \frac{\# \text{ soft unripe} + 1}{\# \text{ soft unripe} + 1 + \# \text{ medium unripe} + 1 + \# \text{ firm unripe} + 1} \\ \mathbb{P}(\text{medium}|\text{unripe}) &\approx \frac{\# \text{ medium unripe} + 1}{\# \text{ soft unripe} + 1 + \# \text{ medium unripe} + 1 + \# \text{ firm unripe} + 1} \\ \mathbb{P}(\text{firm}|\text{unripe}) &\approx \frac{\# \text{ firm unripe} + 1}{\# \text{ soft unripe} + 1 + \# \text{ medium unripe} + 1 + \# \text{ firm unripe} + 1} \end{aligned}$$

*add to*

*1 to numerator can't be 0*

- When smoothing, we add 1 to the count of every group whenever we're estimating a conditional probability.

## Example: Avocados, with smoothing

|   | color        | softness | variety | ripeness |
|---|--------------|----------|---------|----------|
| → | bright green | firm     | Zutano  | unripe   |
| → | green-black  | medium   | Hass    | ripe .   |
| → | purple-black | firm     | Hass    | ripe .   |
| ↓ | green-black  | medium   | Hass    | unripe   |
| ↓ | purple-black | soft →   | Hass    | ripe .   |
| ↓ | bright green | firm     | Zutano  | unripe   |
| ↓ | green-black  | soft →   | Zutano  | ripe .   |
| ↓ | purple-black | soft →   | Hass    | ripe .   |
| ↓ | green-black  | soft ←   | Zutano  | ripe .   |
| ↓ | green-black  | firm     | Hass    | unripe   |
| ↓ | purple-black | medium   | Hass    | ripe .   |

You have a soft green-black Hass avocado. Based on this data, would you predict that your avocado is ripe or unripe?

*Smoothing (for conditional probabilities)*

*$P(\text{ripe} \mid \text{soft}, \text{gb}, \text{Hass})$*

*$\propto P(\text{ripe}) P(\text{soft} \mid \text{ripe}) P(\text{gb} \mid \text{ripe}) P(\text{Hass} \mid \text{ripe})$*

$$\frac{14}{165} = \frac{7}{11} \cdot \frac{4+1}{7+3} \cdot \frac{3+1}{7+3} \cdot \frac{5+1}{7+2}$$

## Example: Avocados, with smoothing

| color        | softness | variety | ripeness |
|--------------|----------|---------|----------|
| bright green | firm     | Zutano  | unripe   |
| green-black  | medium   | Hass    | ripe     |
| purple-black | firm     | Hass    | ripe     |
| green-black  | medium   | Hass    | unripe   |
| purple-black | soft     | Hass    | ripe     |
| bright green | firm     | Zutano  | unripe   |
| green-black  | soft     | Zutano  | ripe     |
| purple-black | soft     | Hass    | ripe     |
| green-black  | soft     | Zutano  | ripe     |
| green-black  | firm     | Hass    | unripe   |
| purple-black | medium   | Hass    | ripe     |

You have a soft green-black Hass avocado. Based on this data, would you predict that your avocado is ripe or unripe?

$$\begin{aligned}
 & P(\text{unripe} | \text{Soft, gb, Hass}) \\
 & \propto P(\text{unripe}) P(\text{Soft} | \text{unripe}) P(\text{gb} | \text{un}) P(\text{Hass} | \text{un}) \\
 & = \frac{4}{11} \cdot \frac{0+1}{4+3} \cdot \frac{2+1}{4+3} \cdot \frac{2+1}{4+2} \\
 & = \frac{4}{11} \cdot \frac{1}{7} \cdot \frac{3}{7} \cdot \frac{3}{6} = \frac{6}{539}
 \end{aligned}$$

Compare

$$\text{ripe} = \frac{14}{165} > \text{unripe}$$

$$\text{unripe} = \frac{6}{539}$$

$\Rightarrow$  would predict a soft, green Hass  $\rightarrow$  ripe

# Summary



## Summary

- In classification, our goal is to predict a discrete category, called a **class**, given some features.

*conditionally independent*

- The **Naïve Bayes** classifier uses Bayes' Theorem:

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

- And works by estimating the numerator of  $\mathbb{P}(\text{class}|\text{features})$  for all possible classes.
- It also uses a simplifying assumption, that features are conditionally independent given a class:

$$\mathbb{P}(\text{features}|\text{class}) = \mathbb{P}(\text{feature}_1|\text{class}) \cdot \mathbb{P}(\text{feature}_2|\text{class}) \cdot \dots$$

Lecture 9 Part 2

# More Naïve Bayes, Review

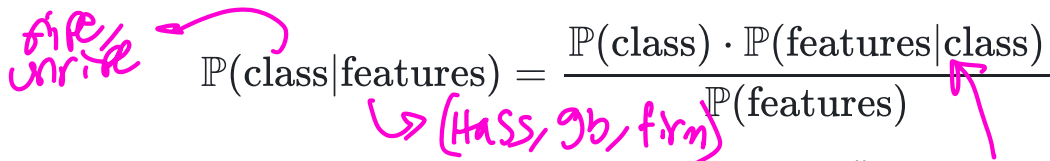
DSC 40A, Summer 2026

## Agenda

- Text classification.
  - Practical demo.
- Review.
  - Old exam problems.

## Recap: The Naïve Bayes classifier

- We want to predict a class, given certain features.
- Using Bayes' Theorem, we write:


$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

- For each class, we compute the numerator using the **naïve assumption of conditional independence of features given the class**.
- We estimate each term in the numerator based on the training data.
- We predict the class with the largest numerator.
  - Works if we have multiple classes, too!

# Text classification

# Text classification

- Text classification problems include:
  - Sentiment analysis (e.g. positive and negative customer reviews).
  - Determining genre (news articles, blog posts, etc.).
- Spam filtering is a common text classification problem:



|                   |                                                  |         |
|-------------------|--------------------------------------------------|---------|
| UPR               | Ultrasonic Pest ReSisSer QQ1Q                    | 4:01 AM |
| HOT97 Summ...     | Last Chance To Celebrate 30 Years Of Summer Jam! | 6/1/24  |
| Koon Thai kit...  | Get 15% OFF At Koon Thai Kitchen                 | 6/1/24  |
| Cori Trattoria... | Thanks San Diego                                 | 5/29/24 |
| Smart H0me...     | Secure Your Home With Vivint                     | 5/29/24 |
| Smart H0me...     | Secure Your Home With Vivint                     | 5/29/24 |
| Pure Barre        | What are the Benefits of Pure Barre?             | 5/29/24 |
| Smart H0me...     | Secure Your Home With Vivint                     | 5/29/24 |

*gmail does this  
everyday*

*industry  
standard*

- **Goal:** Given the body of an email, determine whether it's **spam** or **ham** (not spam).
- **Question:** What information do we use to make these predictions? What features?

## Text features

Idea:

- Choose a **dictionary** of  $d$  words.
- Represent each email with a **feature vector**  $\vec{x}$ :

$$\vec{x} = \begin{bmatrix} x^{(1)} \\ x^{(2)} \\ \dots \\ x^{(d)} \end{bmatrix}_{d \times 1}$$

where  $x^{(i)} = 1$  if word  $i$  is present in the email, and  $x^{(i)} = 0$  otherwise.

This is called the **bag-of-words** model. This model ignores the frequency and meaning of words.

$d=3$  (1) (2) (3)  
"home" "free" "Sale"  
 $\vec{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}_{d \times 1}$   
"come sale your home for sale"  
↑ ↑

does email contain home? → feature

Does not track order of words

$\vec{x}^{(i)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

## Concrete example

• Dictionary: "prince", "money", "free", and "just".

• Dataset of 5 emails (orange are spam, blue are ham):

- 1 ○ "I am the prince of UCSD and I demand money."
- 2 ○ "Tapioca Express: redeem your free Thai Iced Tea!"
- 3 ○ "DSC 10: free points if you fill out SETs!"
- 4 ○ "Click here to make a tax-free donation to the IRS."
- 5 ○ "Free career night at Prince Street Community Center."

1 row/email

features

class

|   | prince | money | free | just | spam? |
|---|--------|-------|------|------|-------|
| 1 | 1      | 1     | 0    | 0    | spam  |
| 2 | 0      | 0     | 1    | 0    | ham   |
| 3 | 0      | 0     | 1    | 0    | ham   |
| 4 | 0      | 0     | 1    | 0    | spam  |
| 5 | 1      | 0     | 1    | 0    | ham   |

How many ways are there to fill in this grid with 1s & 0s?

2 2 22 ... 2  $\Rightarrow 2^{20}$



## Naïve Bayes for spam classification

$$\mathbb{P}(\text{class} \mid \text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features} \mid \text{class})}{\mathbb{P}(\text{features})}$$

- To classify an email, we'll use Bayes' Theorem to calculate the probability of it belonging to each class:
  - $\mathbb{P}(\text{spam} \mid \text{features})$ .
  - $\mathbb{P}(\text{ham} \mid \text{features})$ .
- We'll predict the class with a larger probability.

## Naïve Bayes for spam classification

$$\mathbb{P}(\text{class} \mid \text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features} \mid \text{class})}{\mathbb{P}(\text{features})}$$

- Note that the formulas for  $\mathbb{P}(\text{spam} \mid \text{features})$  and  $\mathbb{P}(\text{ham} \mid \text{features})$  have the same denominator,  $\mathbb{P}(\text{features})$ . *shortcut "proportional to"*

- Thus, we can find the larger probability just by comparing numerators:

- $\mathbb{P}(\text{spam} \mid \text{features}) \propto \mathbb{P}(\text{spam}) \cdot \mathbb{P}(\text{features} \mid \text{spam})$  *4 components*
  - $\mathbb{P}(\text{ham} \mid \text{features}) \propto \mathbb{P}(\text{ham}) \cdot \mathbb{P}(\text{features} \mid \text{ham})$

*How many pieces of info/terms do I need to find/compute to calculate both probabilities?*

*1) pick out events or components first*

## Question 🤔

Answer in chat.

1.  $\mathbb{P}(\text{features} \mid \text{spam})$

2.  $\mathbb{P}(\text{features} \mid \text{ham})$

3.  $\mathbb{P}(\text{spam})$

4.  $\mathbb{P}(\text{ham})$

$p(\text{the} \mid \text{spam}) = \text{pretty big}$   
 $p(\text{the} \mid \text{ham}) = \text{pretty big}$   
1?

$\mathbb{P}(\text{spam})$   $\mathbb{P}(\text{ham})$   $\text{spam} \Leftrightarrow \text{not ham}$   
 $\text{spam} = \overline{\text{ham}}$

Which of these probabilities should add to 1?

~~A. 1, 2~~

☒ B. 3, 4

~~C. Both (a) and (b).~~

~~D. Neither (a) nor (b).~~

## Estimating probabilities with training data

- To estimate  $\mathbb{P}(\text{spam})$ , we compute:

$$\mathbb{P}(\text{spam}) \approx \frac{\# \text{ spam emails in training set}}{\# \text{ emails in training set}}$$

- To estimate  $\mathbb{P}(\text{ham})$ , we compute:

$$\mathbb{P}(\text{ham}) \approx \frac{\# \text{ ham emails in training set}}{\# \text{ emails in training set}}$$

- What about  $\mathbb{P}(\text{features} \mid \text{spam})$  and  $\mathbb{P}(\text{features} \mid \text{ham})$ ?

✓  
conditional  
independence

## Assumption of conditional independence

- Note that  $\mathbb{P}(\text{features} \mid \text{spam})$  looks like:

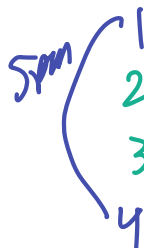
$$\mathbb{P}(x^{(1)} = 0, x^{(2)} = 1, \dots, x^{(d)} = 0 \mid \text{spam})$$

- Recall: the key assumption that the Naïve Bayes classifier makes is that **the features are conditionally independent given the class.**
- This means we can estimate  $\mathbb{P}(\text{features} \mid \text{spam})$  as:

$$\begin{aligned} & \mathbb{P}(x^{(1)} = 0, x^{(2)} = 1, \dots, x^{(d)} = 0 \mid \text{spam}) \\ &= \mathbb{P}(x^{(1)} = 0 \mid \text{spam}) \cdot \mathbb{P}(x^{(2)} = 1 \mid \text{spam}) \cdot \dots \cdot \mathbb{P}(x^{(d)} = 0 \mid \text{spam}) \end{aligned}$$

## Concrete example

- Dictionary: "prince", "money", "free", and "just".
- Dataset of 5 emails (**orange are spam**, **blue are ham**):

- 
- Spam
- 1 ○ "I am the prince of UCSD and I demand money."
  - 2 ○ "Tapioca Express: redeem your free Thai Iced Tea!"
  - 3 ○ "DSC 10: free points if you fill out SETs!"
  - 4 ○ "Click here to make a tax-free donation to the IRS."
  - 5 ○ "Free career night at Prince Street Community Center."

## Concrete example

- New email to classify: "Download a free copy of the Prince of Persia."

|   | prince | money | free | just | spam? |
|---|--------|-------|------|------|-------|
| 1 | 1      | 1     | 0    | 0    | spam  |
| 2 | 0      | 0     | 1    | 0    | ham   |
| 3 | 0      | 0     | 1    | 0    | ham   |
| 4 | 0      | 0     | 1    | 0    | spam  |
| 5 | 1      | 0     | 1    | 0    | ham   |

prince = 1

money = 0

free = 1

just = 0

just the  
numerator  
NOT  
p(spam)

$$\begin{aligned}
 & P(\text{spam} | \text{prince}=1, \text{money}=0, \text{free}=1, \text{just}=0) \propto P(\text{spam}) P(\text{features} | \text{spam}) \\
 & = P(\text{spam}) P(\text{prince}=1 | \text{sp}) P(\text{money}=0 | \text{sp}) P(\text{free}=1 | \text{sp}) P(\text{just}=0 | \text{sp}) \\
 & = \frac{2}{5} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{2}{2} = \frac{1}{20}
 \end{aligned}$$

Does not appear

price = 1  
 money = 0  
 free = 1  
 just = 0

|   | price | money | free | just | spam? |
|---|-------|-------|------|------|-------|
| 1 | 1     | 1     | 0    | 0    | spam  |
| 2 | 0     | 0     | 1    | 0    | ham   |
| 3 | 0     | 0     | 1    | 0    | ham   |
| 4 | 0     | 0     | 1    | 0    | spam  |
| 5 | 1     | 0     | 1    | 0    | ham   |

$$\begin{aligned}
 P(\text{ham} | \text{price}=1, \text{money}=0, \text{free}=1, \text{just}=0) &\propto P(\text{ham}) P(\text{features} | \text{ham}) \\
 &= P(\text{ham}) P(\text{price}=1 | \text{ham}) P(\text{money}=0 | \text{ham}) P(\text{free}=1 | \text{ham}) P(\text{just}=0 | \text{ham}) \\
 &= \frac{3}{5} \cdot \frac{1}{3} \cdot \frac{3}{3} \cdot \frac{3}{3} \\
 &= \frac{1}{5} > \frac{1}{20} \Rightarrow \text{predict ham}
 \end{aligned}$$



Uh oh...

- What happens if we try to classify the email "just what's your price, prince"?

|   | prince | money | free | just | spam? |
|---|--------|-------|------|------|-------|
| 1 | 1      | 1     | 0    | 0    | spam  |
| 2 | 0      | 0     | 1    | 0    | ham   |
| 3 | 0      | 0     | 1    | 0    | ham   |
| 4 | 0      | 0     | 1    | 0    | spam  |
| 5 | 1      | 0     | 1    | 0    | ham   |

features {  
prince = 1  
money = 0  
free = 0  
just = 1

$$p(\text{spam} | \text{features}) \propto p(\text{spam}) p(\text{prince}=1 | \text{spam}) p(\text{money}=0 | \text{spam}) p(\text{free}=0 | \text{spam}) p(\text{just}=1 | \text{spam})$$

0 = ←

↓  
0 spam  
2

## Smoothing

- **Without** smoothing:

$$\mathbb{P}(x^{(i)} = 1 \mid \text{spam}) \approx \frac{\# \text{ spam containing word } i}{\# \text{ spam containing word } i + \# \text{ spam not containing word } i}$$

- **With** smoothing:

$$\mathbb{P}(x^{(i)} = 1 \mid \text{spam}) \approx \frac{(\# \text{ spam containing word } i) + 1}{(\# \text{ spam containing word } i) + 1 + (\# \text{ spam not containing word } i) + 1}$$

- When smoothing, we add 1 to the count of every group whenever we're estimating a conditional probability.

## Concrete example with smoothing

- What happens if we try to classify the email "just what's your price, prince"?

|   | prince | money | free | just | spam? |
|---|--------|-------|------|------|-------|
| 1 | 1      | 1     | 0    | 0    | spam  |
| 2 | 0      | 0     | 1    | 0    | ham   |
| 3 | 0      | 0     | 1    | 0    | ham   |
| 4 | 0      | 0     | 1    | 0    | spam  |
| 5 | 1      | 0     | 1    | 0    | ham   |

$$\begin{aligned}
 & p(s) \cdot p(p=1|s) \cdot p(m=0|s) \cdot p(f=0|s) \cdot p(j=1|s) \\
 &= \frac{2}{5} \cdot \frac{1+1}{2+2} \cdot \frac{1+1}{2+2} \cdot \frac{1+1}{2+2} \cdot \frac{0+1}{2+2} \\
 &= \frac{2}{5} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{4} \\
 &= \frac{1}{80}
 \end{aligned}$$

|   | prince | money | free | just | spam? |
|---|--------|-------|------|------|-------|
| 1 | 1      | 1     | 0    | 0    | spam  |
| 2 | 0      | 0     | 1    | 0    | ham   |
| 3 | 0      | 0     | 1    | 0    | ham   |
| 4 | 0      | 0     | 1    | 0    | spam  |
| 5 | 1      | 0     | 1    | 0    | ham   |

$$\begin{aligned}
 & P(h) P(p=1|h) P(m=0|h) P(f=0|h) P(j=1|h) \\
 & \frac{3}{5} \cdot \frac{1+1}{3+2} \cdot \frac{3+1}{3+2} \cdot \frac{0+1}{3+2} \cdot \frac{0+1}{3+2} \\
 & = \frac{24}{3125} \quad \text{vs} \quad \frac{1}{80} = \frac{40}{3200} \\
 & \quad \text{Bigger}
 \end{aligned}$$

$\Rightarrow$  Spam

expand bag of words model

bigram

generally  $N$ -gram  $\rightarrow$  DSC 80

## Modifications and extensions

- **Idea:** Use pairs (or longer sequences) of words rather than individual words as features.
  - This better captures the dependencies between words.
  - It also leads to a much larger space of features, increasing the complexity of the algorithm.
- **Idea:** Instead of recording whether each word appears, record how many times each word appears.
  - This better captures the importance of repeated words.

That's all the new content we have!

Let's now try out Naïve Bayes in code. Follow along [here](#).

12:47 PT

## Review

We're done with new content! Let's work through some old exam problems.

## Fall 2021 Final Exam, Problem 10

Suppose you're given the following probabilities:

- $\mathbb{P}(A|B) = \frac{2}{5}$ .
- $\mathbb{P}(B|A) = \frac{1}{4}$ .
- $\mathbb{P}(A|C) = \frac{2}{3}$ .

$$P(A|C) = P(A) = \frac{2}{3}$$

Part 1: If  $A$  and  $C$  are independent, what is  $\mathbb{P}(B)$ ?

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)}$$

$$P(B) = \frac{P(A)P(B|A)}{P(A|B)}$$

$$P(B)P(A|B) = P(A)P(B|A)$$

$$\begin{aligned} &= \frac{\frac{2}{3} \cdot \frac{1}{4}}{\frac{2}{5}} \\ &= \frac{1}{6} \cdot \frac{5}{2} = \frac{5}{12} \end{aligned}$$



## Fall 2021 Final Exam, Problem 10

Suppose you're given the following probabilities:

- $\mathbb{P}(A|B) = \frac{2}{5} \cdot \text{pink } P(A) = \frac{2}{5}$
- $\mathbb{P}(B|A) = \frac{1}{4} \cdot \text{pink } P(B) = \frac{1}{4}$
- $\mathbb{P}(A|C) = \frac{2}{3}$

$$P(C) = \frac{P(A)P(C|A)}{P(A|C)} \leftarrow \frac{2}{3}$$

↑

**Part 2:** Suppose  $A$  and  $C$  are not independent, and now suppose that  $\mathbb{P}(A|\bar{C}) = \frac{1}{5}$ .

Given that  $A$  and  $B$  are independent, what is  $\mathbb{P}(C)$ ?

$$P(C) = P(C \cap A) + P(C \cap \bar{A}) = P(A)P(C|A) + P(\bar{A})P(C|\bar{A})$$

$$P(A) = P(A \cap C) + P(A \cap \bar{C}) = P(C)P(A|C) + P(\bar{C})P(A|\bar{C})$$

let  $p = P(C)$

$$\frac{2}{9} = p \cdot \frac{2}{3} + (1-p) \cdot \frac{1}{5} \Rightarrow \text{solve for } p$$

# Announcements

# Announcements

- Homework 8 is due **tomorrow night**. Solutions will not be released so utilize office hours.
- The Final Exam is on **Friday from 11:30AM-2:30PM PT**.
  - You will be emailed a zoom link soon.
  - 160 minutes, on paper, no calculators or electronics, but **you are allowed to bring one double-sided index card (4 inches by 6 inches) of notes that you write by hand.**
- There is a review session **tomorrow from 6-7:30PM PT in the lecture zoom**.
  - Jordan: confirm poll once it closes and will announce the topics
- If you fill out both the **End-of-Term Survey** and **SETs** by 8AM on Friday, then you will have 4 points of extra credit added to your HW7.
- Read the proctoring and formatting guidelines posted on Piazza!

Lecture 9 Part 3

# Review, Final Thoughts

DSC 40A, Summer 2026

## Agenda

- High-level overview of the course.
- Old exam problems.
- Final thoughts.

**What was this course about?**

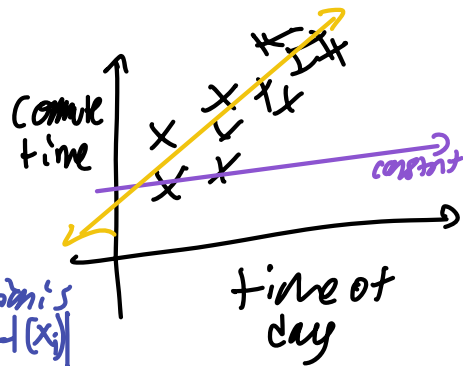
\* finding the best way to make predictions (using data)

## Part 1: Empirical risk minimization (Lectures 1-5)

1. Choose a model.

① constant:  $H(x) = h$  intercept slope

② simple linear regression:  $H(x) = w_0 + w_1 x$



2. Choose a loss function. measures how good or bad a prediction is

④ Squared loss:  $(y_i - H(x_i))^2$   
(actual - predicted)<sup>2</sup>

⑤ Absolute loss  $|y_i - H(x_i)|$

0-1 loss flip loss  
infinitely loss

3. Minimize empirical risk average loss to find optimal model parameters.

$$R_{sq}(h) = \frac{1}{n} \sum_{i=1}^n (y_i - h)^2 \xrightarrow{\text{calculus}} h^* = \text{Mean}(y_1, y_2, \dots, y_n)$$

mean squared error

$$R_{sq}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2 \xrightarrow{\text{calculus}} w_0^* w_1^*$$

→ multiple input variables to make a prediction

multiple linear regression:  $H(\vec{x}) = w_0 + w_1 x^{(1)} + w_2 x^{(2)} + \dots + w_d x^{(d)}$   
 $= \vec{w} \text{Aug}(\vec{x})$

to find  $w_0^*, w_1^*, \dots, w_d^*$

$$\frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x^{(1)} + w_2 x^{(2)} + \dots + w_d x^{(d)}))^2$$

$$X = \begin{bmatrix} \text{Aug}(\vec{x}_1) \\ \text{Aug}(\vec{x}_2) \\ \vdots \end{bmatrix} = \begin{bmatrix} x_1^{(1)} & x_1^{(d)} \\ x_2^{(1)} & x_2^{(d)} \\ \vdots & \vdots \\ x_n^{(1)} & x_n^{(d)} \end{bmatrix} \quad n \times (d+1)$$

$$\vec{w} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{bmatrix} \quad \vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

goal: minimize  $\|\vec{y} - X\vec{w}\|^2$

$\vec{e}$  ← Normal Equations

minimizing  $\vec{e}$  is the same as minimizing MSE

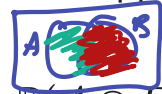


# Why Gradient Descent?

Finding the best way to make predictions  $\rightarrow$  finding best parameters  $\rightarrow$  calculus, lin alg, gradient descent

## Part 2: Probability fundamentals (Lecture 6)

- If all outcomes in the **sample space**  $S$  are equally likely, then  $\mathbb{P}(A) = \frac{|A|}{|S|}$ .
- $\bar{A}$  is the **complement** of event  $A$ .  $\mathbb{P}(\bar{A}) = 1 - \mathbb{P}(A)$ .
- Two events  $A, B$  are **mutually exclusive** if they share no outcomes, i.e. they don't overlap:  $\mathbb{P}(A \cap B) = 0$ .
- For any two events, the probability that  $A$  happens or  $B$  happens is  $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$ .
- The probability that events  $A$  and  $B$  both happen is  $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B|A)$ .
  - $\mathbb{P}(B|A)$  is the probability that  $B$  happens, given that you know  $A$  happened.
  - Through re-arranging, we see that  $\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}$ , which is the definition of conditional probability.



Principle inclusion exclusion

multiplication rule

## Part 2: Combinatorics (Lectures 6-7)

Problem:  
when to apply  
what? subtle

Suppose we want to select  $k$  elements from a group of  $n$  possible elements. The following table summarizes whether the problem involves sequences, permutations, or combinations, along with the number of relevant orderings.

|                                                      | Yes, order matters                               | No, order doesn't matter                           |
|------------------------------------------------------|--------------------------------------------------|----------------------------------------------------|
| <b>With replacement</b><br>Repetition allowed        | $n^k$ possible <b>sequences</b>                  | more complicated: watch <a href="#">this video</a> |
| <b>Without replacement</b><br>Repetition not allowed | $\frac{n!}{(n-k)!}$ possible <b>permutations</b> | $\binom{n}{k}$ <b>combinations</b>                 |

## Part 2: The law of total probability and Bayes' Theorem (Lectures 7-8)

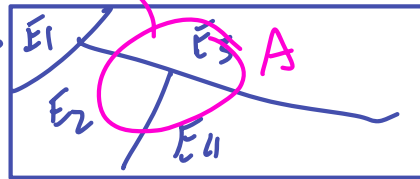
- A set of events  $E_1, E_2, \dots, E_k$  is a **partition** of  $S$  if each outcome in  $S$  is in exactly one  $E_i$ .
- The **law of total probability** states that if  $A$  is an event and  $E_1, E_2, \dots, E_k$  is a partition of  $S$ , then:

$$P(A) = P(A \cap E_1) + P(A \cap E_2) + \dots + P(A \cap E_k)$$

$$P(A) = P(E_1)P(A|E_1) + P(E_2)P(A|E_2) + \dots + P(E_k)P(A|E_k) = \sum_{i=1}^k P(E_i)P(A|E_i)$$

- Bayes' Theorem states that:

$$P(B|A) = \frac{P(B)P(A|B)}{P(A)}$$



Shakeshack in out  
5 guys

new prob

original id

update

- We often re-write the denominator  $P(A)$  in Bayes Theorem' using the law of total probability.

## Part 2: Independence and conditional independence (Lecture 8)

- Two events  $A$  and  $B$  are **independent** when knowledge of one event does not change the probability of the other event.
  - Equivalent conditions:  $\mathbb{P}(B|A) = \mathbb{P}(B)$ ,  $\mathbb{P}(A|B) = \mathbb{P}(A)$ ,  $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ .
- Two events  $A$  and  $B$  are **conditionally independent** given event  $C$  if they are independent given the knowledge that event  $C$  happened.
  - Condition:

$$\mathbb{P}((A \cap B)|C) = \mathbb{P}(A|C)\mathbb{P}(B|C)$$

- In general, there is **no relationship between independence and conditional independence**.
- Make sure you've read **this**!

## Part 2: Naïve Bayes (Lectures 8-9 (today))

- In classification, our goal is to predict a discrete category, called a **class**, given some features.
- The **Naïve Bayes** classifier works by estimating the numerator of  $\mathbb{P}(\text{class}|\text{features})$  for all possible classes.
- It uses Bayes' Theorem:

$$\mathbb{P}(\text{class}|\text{features}) = \frac{\mathbb{P}(\text{class}) \cdot \mathbb{P}(\text{features}|\text{class})}{\mathbb{P}(\text{features})}$$

- It also uses a "naïve" simplifying assumption, that **features are conditionally independent given a class**:

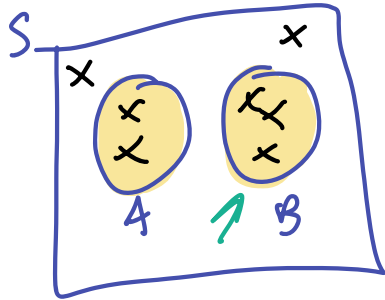
$$\mathbb{P}(\text{features}|\text{class}) = \mathbb{P}(\text{feature}_1|\text{class}) \cdot \mathbb{P}(\text{feature}_2|\text{class}) \cdot \dots$$

# Practice problems

~~★~~ Can you break down the problem for a person not in

## Spring 2023 Midterm Exam 2, Problem 6.2 DSC 40A?

The events  $A$  and  $B$  are mutually exclusive, or disjoint. More generally, for **any** two disjoint events  $A$  and  $B$ , show how to express  $\mathbb{P}(\bar{A} | (A \cup B))$  in terms of  $\mathbb{P}(A)$  and  $\mathbb{P}(B)$  **only**.



What's my  
prob that my  
outcomp is not  
in A?

$$\begin{aligned} P(A | (A \cup B)) &= \frac{P(B)}{P(A \cup B)} \\ &= \frac{P(B)}{P(A) + P(B)} \end{aligned}$$



## Fall 2021 Final Exam, Problem 8

Billy brings you back to Dirty Birds, the restaurant where he is a waiter. He tells you that Dirty Birds has 30 different flavors of chicken wings, 18 of which are 'wet' (e.g. honey garlic) and 12 of which are 'dry' (e.g. lemon pepper).

Each time you place an order at Dirty Birds, you get to pick 4 different flavors. The order in which you pick your flavors does not matter.

**Part 1:** How many ways can we select 4 flavors in total?

$$\binom{30}{4}$$

**Part 2:** How many ways can we select 4 flavors in total such that we select an equal number of wet and dry flavors?

$$\binom{18}{2} \cdot \binom{12}{2}$$

$\frac{4}{2} = 2$  per wet/dry  
AND  $\rightarrow$  at the same time  $\rightarrow$  multiply

## Fall 2021 Final Exam, Problem 8

Billy brings you back to Dirty Birds, the restaurant where he is a waiter. He tells you that Dirty Birds has 30 different flavors of chicken wings, 18 of which are 'wet' (e.g. honey garlic) and 12 of which are 'dry' (e.g. lemon pepper).

Each time you place an order at Dirty Birds, you get to pick 4 different flavors. The order in which you pick your flavors does not matter.

**Part 1:** How many ways can we select 4 flavors in total?

**Part 2:** How many ways can we select 4 flavors in total such that we select an equal number of wet and dry flavors?

(

**Part 3:** Billy tells you he'll surprise you with 4 different flavors, randomly selected from the 30 flavors available. What's the probability that he brings you at least one wet flavor and at least one dry flavor?

$$\frac{\text{\# combinations w/ at least 1 wet \& 1 dry flavor}}{\text{\# combinations of 4 flavors}}$$

$$\frac{\binom{30}{4} - \binom{18}{0}\binom{12}{4} - \binom{18}{4}\binom{12}{0}}{\binom{30}{4}}$$

possible combinations

① complement

$$\text{\# total} - \text{\# no wet} - \text{\# no dry}$$

$$\binom{30}{4} - \binom{18}{0}\binom{12}{4} - \binom{18}{4}\binom{12}{0}$$

→  
= 1 way to  
choose NO wet wings

② directly

1w 3d  
2w 2d  
3w 1d

$$\binom{18}{1}\binom{12}{3} + \binom{18}{2}\binom{12}{2} + \binom{18}{3}\binom{12}{1}$$

1 - smth, complement

**Part 4:** Suppose you go to Dirty Birds once a day for 7 straight days. Each time you go there, Billy brings you 4 different flavors, randomly selected from the 30 flavors available. What's the probability that on **at least one** of the 7 days, he brings you all wet flavors or all dry flavors? (Note: All 4 flavors for a particular day must be different, but it is possible to get the same flavor on multiple days.)

☐ at least once  $\xrightarrow{\text{complement}}$  ☐ never happens

☐ never happens if (never get all wet or all dry flavors)

$$\begin{aligned}
 &= P(\text{don't get all wet/all dry day 1}) \times P(\dots \text{day 2}) \times \dots \text{last slide} \\
 &= P(\text{"day 1"})^7 = P(\text{at least 1 wet, 1 dry})^7 \\
 &= 1 - P(\text{at least 1 wet 1 dry day 1})^7 = 1 - \left( \frac{\binom{30}{4} - \binom{18}{4} - \binom{12}{4}}{\binom{30}{4}} \right)^7
 \end{aligned}$$

## Fall 2021 Final Exam, Problem 9

In this question, we'll consider the phone number 6789998212 (mentioned in Soulja Boy's 2008 classic, "Kiss Me thru the Phone").

Part 1: How many permutations of 6789998212 are there?

Handwritten solution for Part 1:

10! (Total permutations of 10 digits)  
 3! 2! 2! (Adjustments for repeated digits: 3 9s, 2 8s, 2 2s)  

$$= \frac{10!}{3! 2! 2!}$$

Alternative calculation using combinations:

10 positions, 3 of them are 9s so choose 3 of those 10 spots  

$$= \binom{10}{3} \binom{7}{1} \binom{6}{2} \binom{4}{1} \binom{3}{2} \binom{1}{1}$$

Order of bottom values doesn't really matter will cancel out

Part 2: How many permutations of 6789998212 have all three 9s next to each other?

Handwritten solution for Part 2:

8! (8 characters total left)  
 2! 2! (Adjustments for repeated digits: 2 8s, 2 2s)  

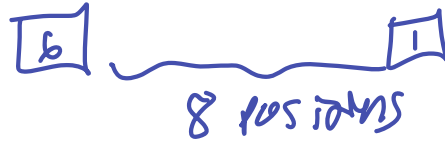
$$= \frac{8!}{2! 2!}$$

8's + 2's are repeated but 88 = 88 so remove the double counts

3 9s

**Part 3:** How many permutations of 6789998212 end with a 1 and start with a 6?

$$\frac{8!}{2! \cdot 3! \cdot 2!}$$



$$\begin{matrix} 7 \\ 8 \ 8 \\ 9 \ 9 \ 9 \\ 2 \ 2 \end{matrix}$$

**Part 4:** How many different 3 digit numbers with unique digits can we create by selecting digits from 6789998212? *→ easier ten involved, → 2 0s but here*

$$\frac{6}{1} \cdot \frac{5}{1} \cdot \frac{4}{1} = 120$$

$$\binom{6}{3} 3!$$

*need 3! b/c the order matters*

*of 6 digits, pick 3 of them ← only gets a set*

$$\begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} \begin{matrix} 6 \\ 7 \\ 8 \\ 9 \\ 1 \\ 2 \end{matrix}$$

# Final thoughts

## Learning objectives

On the first day of the quarter, we told you that after taking DSC 40A, you would:

- understand the basic principles underlying almost every machine learning and data science method.
- be better prepared for the math in upper division: calculus, linear algebra, and probability.



## What's next?

In DSC 40A, we just scratched the surface of the theory behind data science. In future courses, you'll build upon your knowledge from DSC 40A, and will learn:

- More **supervised learning**, e.g. logistic regression, decision trees, neural networks.
- **Unsupervised learning**, e.g. clustering, PCA.
- More **probability**, e.g. random variables, distributions, stochastic processes.
- More **connections** between all of these areas, e.g. the relationship between probability and linear regression.
- More practical tools.

↳ DSC 80 → implementation of the theory

↖ DSC 140A

# Thank you!

This course would not have been possible without our TA:

Eric Ness

[eness@ucsd.edu](mailto:eness@ucsd.edu)

Congrats on (almost) finishing DSC 40A!  
Good luck on the final, and **please keep in touch!**

[j6schwartz@ucsd.edu](mailto:j6schwartz@ucsd.edu)

[jjschwa@engineering.upenn.edu](mailto:jjschwa@engineering.upenn.edu)